

Our Design Thinking Journey: Understanding and Addressing the Hyderabad Housing Crisis

Introduction

Every successful project begins with a group of individuals who share a common curiosity and desire to solve meaningful problems. As a team, we all enjoyed research, analysis, critical thinking, and collaborative problem-solving. When the project began, we explored several potential themes and spent considerable time discussing which problem space would allow us to create the greatest impact.

After multiple meetings and discussions, we selected the theme **Infrastructure & Socio-Economic Stability**. This theme focuses on tangible societal challenges such as housing affordability, transportation accessibility, and the financial pressures experienced by individuals in an increasingly complex and automated economy.

Our decision was largely influenced by our personal observations and experiences. Many of us had witnessed the challenges associated with purchasing or renting homes and the financial burdens that often accompany housing decisions. We recognized that housing is not merely a physical necessity but a deeply interconnected issue that affects economic security, quality of life, and social stability.

With a shared interest in understanding these challenges, we began our journey.

Initial Research and Exploration

The first phase of our project involved extensive secondary research. We explored research papers, news articles, government reports, policy studies, housing market analyses, and case studies to better understand the housing ecosystem.

Through our research, we discovered several recurring issues within the housing sector, including:

- Artificial scarcity of housing supply
- Housing price inflation
- Corruption and inefficiencies within housing allocation systems
- Information asymmetry between buyers, renters, and authorities
- Financial barriers to home ownership
- Challenges faced by migrants and informal workers
- Lack of transparency in housing-related processes

As our research progressed, we realized that these problems were not isolated. Each issue influenced several others, creating a complex web of interconnected challenges.

To make sense of the information, we began mapping relationships between problems. This helped us visualize how housing affordability, financial stress, urban planning, migration, and governance interacted with one another. Rather than viewing housing as a standalone issue, we started understanding it as part of a larger socio-economic ecosystem.

This stage provided us with a strong foundation and prepared us for direct engagement with users.

The Challenge of Understanding Real People

While secondary research gave us valuable information, we understood that meaningful solutions could only emerge by listening to people directly affected by these issues.

Our objective was to speak with individuals who had experience purchasing homes, renting properties, or dealing with housing-related financial challenges.

However, our first attempts at conducting interviews were not successful.

Many participants were hesitant to discuss their financial situations, housing struggles, or personal vulnerabilities. Questions regarding income, affordability, loans, and financial stress often resulted in short or incomplete responses.

Initially, this was frustrating. We had prepared questions carefully and expected more detailed conversations. However, after reflecting on the situation, we realized that the issue was not with the participants but with our approach.

We were approaching interviews as information-gathering exercises rather than human conversations.

This realization became one of the most important turning points in our project.

We redesigned our interview strategy by introducing ice-breakers, adjusting the language we used, and tailoring conversations to individual participants. Instead of immediately asking sensitive questions, we focused on building trust and creating a comfortable environment.

We learned that empathy was not simply a design-thinking tool; it was the foundation of meaningful communication.

Once participants felt comfortable, they became significantly more open and willing to share their experiences.

Data Collection

With our improved interview strategy, we conducted field interviews and interacted with people from different backgrounds and circumstances.

These conversations provided valuable insights into the realities of housing affordability, renting difficulties, financial planning, documentation challenges, and interactions with housing systems.

After collecting responses, we compiled all interview data and generated transcripts for further analysis.

This marked the transition from gathering information to understanding patterns.

Data Analysis and Persona Development

The next phase involved a detailed analysis of our collected data.

We carefully examined interview transcripts and conducted personality and behavioral analysis to identify common characteristics among participants.

Based on these observations, we created personas that represented different categories of users. These personas helped us understand how different groups experience housing-related challenges in unique ways.

Several patterns began to emerge:

- Housing affordability challenges varied significantly across income groups.
- Housing choices were heavily influenced by locality and access to infrastructure.
- Renting difficulties often depended on awareness and technological literacy.
- Migrant workers faced additional documentation and eligibility barriers.
- Financial insecurity affected decision-making across all user groups.

The persona development process allowed us to move beyond individual stories and identify larger trends within the housing ecosystem.

Defining the Problem

Although our theme covered multiple socio-economic issues, our research and interviews consistently pointed toward housing as a major area of concern.

After extensive discussions and review sessions, we decided to narrow our focus and address the **Housing Crisis in Hyderabad**.

This decision allowed us to concentrate our efforts on a specific problem while still addressing broader socio-economic challenges.

Defining the problem was not a straightforward task. In many ways, it was more difficult than solving it. We spent significant time debating which aspect of the housing crisis would provide the most meaningful opportunity for intervention.

Eventually, we agreed on a focused direction and moved forward.

Challenges and Failures Along the Way

Although the final narrative of a project often appears smooth and linear, our journey was filled with obstacles, setbacks, and moments of uncertainty.

Information Overload

As research progressed, we accumulated an enormous amount of information. At one point, the volume of data became overwhelming.

We had articles, reports, interview transcripts, observations, notes, and discussions pointing toward dozens of different problems.

Rather than clarifying the situation, the abundance of information initially made decision-making more difficult.

To overcome this challenge, we began categorizing information and mapping relationships between issues. This transformed scattered observations into structured knowledge.

Interview Setbacks

Our first interviews were not effective because participants were uncomfortable sharing sensitive information.

This forced us to completely rethink our interviewing process and taught us the importance of empathy and trust-building.

Difficulty Narrowing the Scope

Every problem we investigated seemed connected to several others.

The deeper we researched, the more challenges we discovered. Housing affordability connected to transportation, transportation connected to employment, employment connected to financial stability, and so on.

Limiting our focus became one of the hardest parts of the project.

Team Discussions and Conflicting Opinions

As with any collaborative project, team members often had different perspectives regarding priorities, interpretations, and solution directions.

These disagreements occasionally led to lengthy discussions.

However, these conversations ultimately strengthened the project by forcing us to justify our decisions and consider alternative viewpoints.

Journey Mapping and Root Cause Analysis

Before generating solutions, we wanted to ensure that we fully understood the underlying causes of user frustrations.

Using our personas, we created detailed journey maps that illustrated each user's experiences throughout their housing journey.

These maps helped us identify pain points, moments of frustration, and areas where users encountered barriers.

For every major pain point, we conducted root cause analyses.

Rather than focusing only on symptoms, we repeatedly asked deeper questions to identify underlying causes.

This process improved our understanding of the problem and prevented us from developing superficial solutions.

Ideation

Once we had developed a comprehensive understanding of the housing crisis, we moved into the ideation phase.

For every "How Might We" question, we conducted structured brainstorming sessions.

Our process consisted of multiple rounds:

1. Each team member generated three original ideas.
2. The next team member expanded upon those ideas.
3. Additional rounds continued building on previous contributions.

This process encouraged creativity, collaboration, and collective thinking.

As a result, we generated hundreds of ideas spanning multiple areas of the housing ecosystem.

Our ideas covered themes such as:

- Housing affordability
- Tenant and landlord trust

- Transparent housing allocation systems
- Migrant worker inclusion
- Financial resilience and EMI management
- Urban planning and infrastructure transparency

The scale of ideation exceeded our expectations and demonstrated the diversity of perspectives within the team.

The Challenge of Convergence

Generating ideas turned out to be much easier than selecting them.

By the end of our ideation sessions, we had developed hundreds of concepts spread across multiple ecosystems and categories.

At first, this felt like a success. However, it quickly became clear that implementing all of these ideas was impossible.

Many ideas were desirable but difficult to implement. Others were technically feasible but created limited value. Some addressed symptoms rather than root causes.

This forced us to make difficult decisions.

We evaluated every idea using criteria such as:

- Desirability
- Feasibility
- Viability
- Implementation effort
- Potential impact

We used discussions, prioritization frameworks, and value-effort analysis to compare alternatives objectively.

Many ideas that initially seemed promising were eventually discarded.

Although challenging, this process taught us that innovation is not only about generating ideas but also about selecting the right ones.

Sorting and Refinement

After evaluation, we began searching for common themes across ideas.

Many solutions addressed similar problems from different perspectives. Instead of treating them as separate concepts, we combined related ideas into broader solution ecosystems.

This process allowed us to reduce redundancy and identify stronger, more comprehensive concepts.

Eventually, we consolidated our work into three major prototype directions.

Each prototype incorporated multiple ideas and addressed different aspects of the housing crisis.

Selecting the Final Prototype

After comparing the prototypes, we selected the solution that best balanced user needs, impact, feasibility, and scalability.

Our chosen concept focused on improving transparency, accessibility, and trust within housing allocation systems.

The prototype included features such as:

- Housing scheme eligibility guidance
- Application assistance
- Transparent status tracking
- SMS-based updates
- Live queue monitoring
- Mobile service appointments
- Waitlist transparency
- Guided rejection recovery processes

The objective was not merely to digitize an existing process but to reduce confusion, improve accessibility, and increase trust among applicants.

Key Learnings

Throughout this journey, our greatest lessons emerged from the challenges we faced.

We learned that:

- Empathy is essential for understanding users.
- Defining the right problem is often harder than solving it.
- Research without organization can create confusion rather than clarity.
- Not every good idea should be pursued.
- Collaboration requires patience, communication, and compromise.
- Failure and iteration are necessary parts of innovation.
- Understanding root causes leads to stronger solutions.

Conclusion

This project was far more than a study of housing challenges. It was a journey of research, empathy, critical thinking, collaboration, and problem-solving.

From selecting a theme and conducting extensive research to interviewing users, analyzing data, identifying root causes, generating hundreds of ideas, overcoming failures, and converging on a meaningful solution, every stage contributed to our growth as a team.

Most importantly, the project taught us that impactful solutions emerge not from assumptions, but from a deep understanding of the people affected by the problem.

By the end of this journey, we not only developed a solution for addressing aspects of the housing crisis in Hyderabad but also gained invaluable experience in approaching complex real-world challenges through empathy, structure, and innovation.